

Stage 2 Specialist Mathematics

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Subject Co-ordinator	R Scott
Teaching Staff	M Durant, R Scott
Texts	<i>Mathematics for Australia- 12 Specialist Mathematics; Haese et al</i>
Lesson Allocation	Five 45 or 50 minute lessons per week

Rationale

Specialist Mathematics draws on and deepens students' mathematical knowledge, skills, and understanding, and provides opportunities for students to develop their skills in using rigorous mathematical arguments and proofs, and using mathematical models. It includes the study of functions and calculus.

The subject leads to study in a range of tertiary courses such as mathematical sciences, engineering, computer science, and physical sciences. Students envisaging careers in related fields will benefit from studying this subject.

Specialist Mathematics is designed to be studied in conjunction with Mathematical Methods.

Learning Requirements

The learning requirements summarise the key skills, knowledge and understanding that students are expected to develop and demonstrate through learning in Stage 2 Mathematical Methods.

In this subject, students are expected to:

1. understand mathematical concepts, demonstrate mathematical skills, and apply mathematical techniques
2. investigate and analyse mathematical information in a variety of contexts
3. think mathematically by posing questions, solving problems, applying models, and making, testing, and proving conjectures
4. interpret results, draw conclusions, and determine the reasonableness of solutions in context
5. make discerning use of electronic technology to solve problems and to refine and extend mathematical knowledge
6. communicate mathematically and present mathematical information in a variety of ways.

These learning requirements form the basis of the:

- learning scope
- evidence of learning that students provide
- assessment design criteria
- levels of achievement described in the performance standards.

Term 1	Content	Text Reference	Assessment
Week 1 Lessons commence Thursday (TBC)	Mathematical Induction Sums, products and divisibility Including polynomials, matrices, trig functions and calculus [additional handout may replace textbook questions] No need for summation & product notation. No need for inequalities and recursive functions	1B (not Q4,6,7, 8,11c,13)	
Week 2 Wed: SS photos Thursday morning: Commencement of Year Ceremony	Induction continued. Polynomials: Operations, Roots, zeros and factors	2A, B	
Week 3	Real Polynomials: Equality, Equating coefficients, Definition, $P = dQ + R$, Division, Synthetic Division (optional), The Remainder theorem	2C, D, E	
Week 4 Tu-Fri: 12IB and SACE Geography KI Trip	The Factor theorem, Factorisation of $x^n \pm a^n$, The Fundamental theorem of Algebra	2F, G	
Week 5 Sun-Tues: 12 Ancient Studies to Melbourne Trip	Graphing real polynomials, Polynomial Equations	2I (briefly), J	SAT 1: Induction and Polynomials
Week 6 Mon-Wed: Yr12 Outdoor Ed Camp 1 Wed-Fri: Yr12 Outdoor Ed Camp 2	Complex Numbers: Complex Plane, Modulus and Argument,	<i>"Exploration of solutions to $z^n = w$"</i> <i>"Operations with $\mathbb{C}$" Worksheet</i> 4A, 4B & Exercise on p107 of electronic text, C1	
Week 7 Mon: Adelaide Cup Day Tues: Sport photos MS & SS Tues: IGSSA Swimming Thur L3to6 – Yr11&12 SACE Drama & IB Theatre Excursion Weekend: Head of the River	Polar form, argument, Multiplying and dividing in Polar form, argument properties, Conversion between Cartesian and Polar form, Locus problems	4C2, C3, 4D (ext) Additional worksheets for drawing loci	
Week 8 Mon-Fri: Yr11 Outdoor Ed Camp Group 1	De Moivre's Theorem, Solving $z^n = C$, the n^{th} roots of unity,	4E, 4F	
Week 9 Summer InterSchool Week Mon-Fri: Yr11 Outdoor Ed Camp Group 2 Wed-Thur: Yr 11 Tourism Barossa Trip	Catch up & revision Functions: Composites, Inverse functions	3A, 3B	SAT 2: Complex Numbers
Week 10 Fri: Jazz Cabaret Rehearsal Day Sat: SS Formal	Reciprocal and rational functions	3C, 3D, 3E + <i>"Graphs of Rational Functions" Worksheet</i>	
Week 11 Fri: 12:15pm finish	Absolute value functions (NB: Text does not cover $ f(x) $.)	<i>"Modulus Functions" worksheet</i> 3F	

Term 2	Content	Text Reference	Assessment
Week 1 Mon: Staff PD day 12 IB Mini-Exam Week Fri, Sat, Sun: Generations in Jazz	3D Vectors: Vectors in space, Operations with Vectors, Vector algebra, Coordinate geometry of vectors, Parallelism,	5A, 5B, 5C 5D, 5E	
Week 2	Collinearity and unit vectors Scalar product and algebraic properties, Scalar product geometric properties, Vector proof Vector product definition and properties	5F, 5G, 5H + "Vector Projection" & "Vector Proof Suppl Qns" 5I	
Week 3 Tues: MS Swimming Carnival Thur-Sat: Yr 12 Art Melbourne Trip	Integration: Integration Rules Inverse Trig Integrals, Integration by Substitution	7A 7B, C1 (not Q10,11 ext) C2	SAT 3: Functions and Vectors I
Week 4	Integration by Parts (LIATE approach?) Areas, Solids of revolution	7D, E 7F	
Week 5 Tues: Year 11 Retreat Day (morning) Thurs: SS cross country at lunch	Catchup & Test Rates of Change and D.E's: Implicit Differentiation, Related rates	8A, B	
Week 6	Differential Equations Separable D.E's	8C (Q1 &2) 8D, 8E	SAT 4: Integration & Related Rates (not including implicit differentiation)
Week 7 Mon: PH King's B'Day Tues: 11 IB Study Day at home Tues: 11 SACE Study Day at home W-F: 11 IB and 11 SACE Exams Th-Sat: 12 SACE Drama Performances	Slope fields, Problem-solving	<i>"Intro to Slope Fields" wksht</i> + 8F, G,	
Week 8 Mon: 11 IB Study Day at home Mon: 11 SACE Study Day at home Tues-Fri: Yr12 Outdoor Ed Flinders Trip T-Th: 11 IB and 11 SACE Exams	Logistic growth, Parametric equations	8H, 9A	
Week 9 Work Experience Week Mon-Fri: Yr11 SACE We-Fri: Yr 11 IB and Accelerated Friday: Final Full Day	Investigation		Start Investigation

Term 3	Content	Text Reference	Assessment
Week 1 Mon: Students return	Pairs of uniformly varying quantities, Pairs of non-uniformly varying quantities	9B, 9C	
Week 2 M-F: Yr11 SACE Drama & Yr11 IB Theatre Incursion Week W-F: Yr 12 Tourism Trip (TBC)	Bezier Curves (optional), Circular and Elliptical Motion, Arc length	9D (Q2-4), 9E + extra, 9F	Investigation Draft Part 1 Due
Week 3 Tues: SS Winter Sport Photos Fri: Yr12 Valet Photo Day	Investigation		
Week 4 Winter InterSchool Week Tues-Fri: ABODA music festival	Review + Test Vectors: Lines and Planes Area, Lines in 2D and 3D	6A, B	SAT 5: Rates of Change, DE's and Vector Calculus (including implicit differentiation)
Week 5 Winter InterSchool Week (ctd?) Tues-Fri: Yr12 Outdoor Ed Flinders Trip	Angle between two lines, Constant velocity problems, Distance of a point to a line, Intersecting lines, Relationships between lines,	6C, D, E 6F, 6G	Investigation Draft Part 2 Due (TBC)
Week 6	Cartesian Eqn of a plane	6H	
Week 7 Monday: Yr12 IB Study Day Mon-Tues: Yr12 SACE Performances Tues-Thurs: Y12 IB Mock Exam Week Friday: Student Free Day	Angles in Space, Row operations, Intersecting planes	6I, J, K	
Week 8 Mon-Fri: Yr12 SACE & IB Mock Exam Week	Mock Exams		
Week 9 Year 12 Classes resume Tues: MS/SS Music Photos Thurs-Sat: Musical Showcase	Investigation Final + Submit		Investigation Final Due (TBC)
Week 10 Wed: IGSSA Athletics Day Thurs: Paul Dillion talks (TBC) Thurs: Achilles Cup	Catch-up, revision and test		SAT 6: Vector II Lines & Planes

Term 4	Content	Text Reference	Assessment
Week 1 Mon: Students return Wed: House Athletics Day Thurs: Valedictory Rehearsals Fri: Valedictory Ceremony & Yr 11 Study Day	Leavers Week		
Week 2 Tues: Y12 IB Exams start Thu- Fri: Yr11 Outdoor Ed Kayaking Days	Exam Revision (Lesson's at teachers' discretion)		
Week 3 Tues-Fri: Yr11 Outdoor Ed Renmark Trip	Exam Revisions		
Week 4	Exams commence		

ASSESSMENT

The mark required for SACE assessment will be calculated as follows:

Assessment type	Number of tasks	Total Weighting
Skills and Applications Tasks	SAT 1: Induction & Polynomials – 1 A4 page of notes and calculator permitted SAT 2: Complex Numbers – 1 A4 page of notes and calculator permitted SAT 3: Functions and Graphs & Vectors I – 1 A4 page of notes and calculator permitted SAT 4: Integration, Implicit Differentiation and Related Rates – 1 A4 page of notes and calculator permitted SAT 5: Vector Calculus – 1 A4 page of notes and calculator permitted SAT 6: Vectors II: Lines and Planes – 1 A4 page of notes and calculator permitted	50%
Mathematical Investigation	1 task – maximum 12 pages	20%
Examination (130 minutes)	External Two double sided A4 notes pages and calculator permitted	30%

- The six tests will be approximately equal in weighting.
- The mock examination in Term 3 will be for Pembroke assessment and will aid in the calculation of the predicted score.